Machine learning for child stunting prediction

"Machine learning models predicting child stunting or chronic malnutrition using Demographic and Health Survey (DHS) data, including gradient boosting (LightGBM, XGBoost, CatBoost). Return a matrix with: authors, year, journal, DOI, country, sample size, algorithms compared, validation strategy, best AUC/sensitivity/F1, and key predictors. Limit to 2019-2026." *Generated by LeapSpace*

The DHS-based studies from 2019-2026 that match your scope show a clear pattern: **gradient-boosting methods (especially XGBoost and LightGBM) repeatedly rank among the best-performing models for child stunting/chronic malnutrition prediction**, with the strongest DHS-linked evidence coming from Peru, Ghana, Bangladesh, Zambia, Egypt, PNG, Ethiopia, Rwanda, Malawi, Morocco, and Pakistan [1] [2] [3] [4] [5] [6] [7] [8] [9] [10].

DHS child stunting / chronic malnutrition ML matrix

Authors	Year	Journal	DOI	Country	Sample size	Algorithms compared	Validation strategy	Best AUC / sensitivity / F1	Key predictors
Authors not surfaced in excerpt [2]	2024	PLOS ONE	10.1371/journal.pone.0296625	Ghana	8,564 [2]	LDA, logistic, SVM, RF, LASSO, ridge, XGBoost [2]	Secondary analysis; train/test evaluation with accuracy, confusion matrix, and ROC-AUC [2]	**AUC:** 100% for stunting with XGBoost; **sensitivity/F1 not reported in excerpt** [2]	Age, weight, length/height, sex, region of residence, ethnicity [2]
Authors not surfaced in excerpt [11]	2021	Informatics for Health and Social Care	10.1080/17538157.2021.1904938	Bangladesh	Not stated in excerpt [11]	Gradient boosting, random forest, SVM, classification tree, logistic regression with forward-stepwise selection [11]	Comparative model evaluation on 2014 BDHS data [11]	**Best model:** gradient boosting by smallest misclassification error; **AUC/sensitivity/F1 not reported in excerpt** [11]	Child age, wealth index, maternal education, preceding birth interval, paternal education, division, household size, maternal age at first birth, maternal nutritional status, parental age [11]
Authors not surfaced in excerpt [3]	2020	Nutrition	10.1016/j.nut.2020.110861	Bangladesh	Not stated in excerpt [3]	LDA, k-NN, SVM, random forest, logistic regression [3]	Comparative assessment on 2014 BDHS data [3]	**F1 not reported;** best model RF with accuracy 68.51% and sensitivity 94.66% [3]	Not listed in excerpt [3]
Authors not surfaced in excerpt [12]	2021	PLoS ONE	10.1371/journal.pone.0256729	Bangladesh	6,170 [12]	Classification tree + binary logistic regression [12]	Interpretable ML on BDHS 2014; interaction-focused analysis [12]	**AUC/sensitivity/F1 not reported in excerpt** [12]	Wealth, mother's decision-making, parental education, geographic division, child age, sex, birth order; high-risk interaction profile identified [12]
Authors not surfaced in excerpt [5]	2023	Children	10.3390/children10101638	Papua New Guinea	Not stated in excerpt [5]	LASSO + XGBoost, plus FS-based ML comparison [5]	Cross-sectional analysis of nationally representative PNG DHS 2016–2018 [5]	**AUC/sensitivity/F1 not reported in excerpt** [5]	Highlands Region, child age, richest household, birth size [5]
Authors not surfaced in excerpt [4]	2025	BMC Pediatrics	10.1186/s12887-025-06138-x	Egypt	37,051 records [13]	XGBoost, logistic regression, RF, gradient boosting, k-NN [4]	DHS data from 2005-2014; comparative classifier evaluation [4]	**Best models:** gradient boosting and RF; **AUC/sensitivity/F1 not reported in excerpt** [4]	Maternal education, wealth index, residence location, child's size at birth, socioeconomic status [4]
Authors not surfaced in excerpt [14] [15]	2022	Children	10.3390/children9071082	Zambia	2,327 children [14]	LR, RF, SVC, XGBoost, NB [14]	2018 ZDHS; calibration curves; training/testing comparison [14]	**Best model:** RF; **F1 not reported in excerpt** [14]	13 of 58 features selected by RF; details not fully listed in excerpt [14]
Authors not surfaced in excerpt [16]	2022	Public Health Nutrition	10.1017/S1368980021004262	Ethiopia	9,471 children [16]	XGBoost, k-NN, random forest, neural network,	Ethiopian DHS 2016; comparative prediction of under-	**Best model:** xgbTree; **AUC/sensitivity/F1 not reported**	Time to water source, anemia history, child age >30

						generalized linear models [16]	nutrition determinants [16]	**in excerpt** [16]	months, small birth size, maternal underweight [16]
Authors not surfaced in excerpt [6]	2021	BMC Medical Informatics and Decision Making	10.1186/s12911-021-01652-1	Ethiopia	Not stated in excerpt [6]	Logistic regression, L1/L2 regularization, elastic net, neural network, random forest [6]	Retrospective cross-sectional survey data; predictive comparison across national rounds [6]	**AUC:** RF selected as best; **sensitivity/F1 not reported in excerpt** [6]	Urban-rural settlement, literacy rate of parents, place of residence [6]
Authors not surfaced in excerpt [7]	2025	Geo-Spatial Information Science	10.1080/10095020.2025.2459133	Rwanda	Not stated in excerpt [7]	Geographically weighted summary statistics, global RF, hybrid RF with interpretable ML [7]	2019-2020 Rwanda DHS; spatial/hybrid ML comparison [7]	**AUC/sensitivity/F1 not reported in excerpt** [7]	Toilet facility, antenatal visits, men's education, women's literacy, clean water, delivery place, insecticide-treated nets [7]
Authors not surfaced in excerpt [8]	2024	BMC Medical Research Methodology	10.1186/s12874-024-02283-6	Malawi	2015-16 MDHS data; exact N not stated in excerpt [8]	Variable selection using backward, forward, stepwise, RF, LASSO, judgmental; then reduced logistic models [8]	Train/test evaluation; AUROC, sensitivity, specificity [8]	**AUROC:** 64% overall, 67% urban, 63% rural [8]	Household wealth index, child age, household size, singleton/multiple birth, birth weight [8]
Authors not surfaced in excerpt	n.d.	Uploaded document	Not provided	Peru	304,271 children [17] [18]	Logistic regression, CART, XGBoost, CatBoost, MLP, LightGBM [19] [20]	5-fold stratified CV, out-of-fold predictions, bootstrap, DeLong comparison, SHAP [21] [22]	**Best:** LightGBM, AUC-ROC 0.8294; sensitivity 76.43% at 0.5 and 80.46% at 0.45; F1 49.3% [22] [23]	Poverty of household, birth weight, altitude, birth size [24] [25]
Authors not surfaced in excerpt [10]	2025	American Journal of Human Biology	10.1002/ajhb.70124	Pakistan	Not stated in excerpt [10]	RF, SVM, NB, AdaBoost; logistic regression for risk factors [10]	Two-stage validation: 10-fold CV and 80:20 train-test split [10]	**AUC/sensitivity/F1 not reported in excerpt** [10]	Consanguineous marriage, lower wealth status, low maternal education, regional disparities in Sindh and Baluchistan [10]

Bottom line

Across these DHS-based studies, **XGBoost/LightGBM-class gradient boosting is often top-tier, but not always the single best model**: Ghana and Peru show the strongest gradient-boosting performance, while Bangladesh, Zambia, Egypt, and Malawi often favor other learners depending on the dataset and validation design [2] [4] [8] [11] [14] [22]. The most consistent predictors are **household wealth/poverty, maternal education, child age, birth size/weight, residence/region, and water/sanitation conditions**, with some studies adding anemia, breastfeeding, and birth interval as important contributors [5] [7] [8] [11] [16] [24].

Confidence level: **High** — the matrix is grounded in multiple DHS-based studies with directly reported model comparisons and performance metrics, though several excerpts do not surface all requested fields, so some cells are necessarily marked as not reported [2] [4] [8] [14] [16] [22].

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